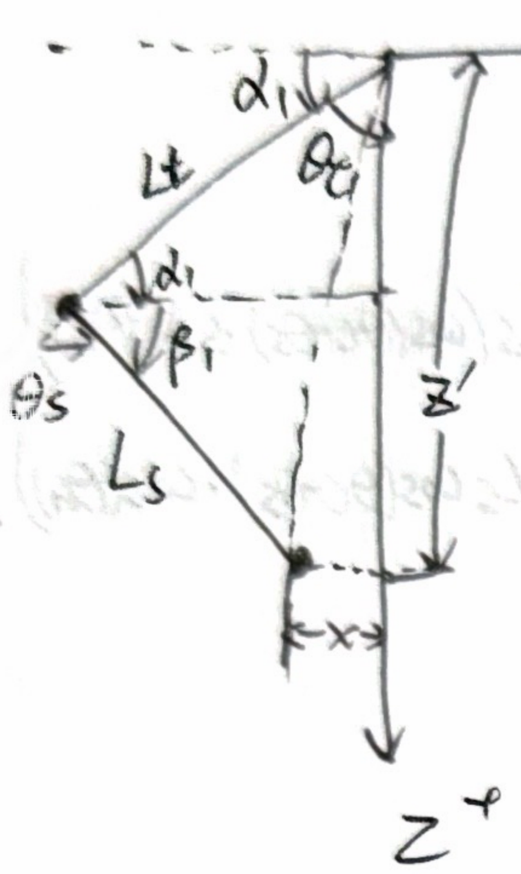
Quadruped Robot Jacobian Derivation:

1. Planar FK:



$$\sin\left(\alpha+\frac{\pi}{2}\right)=\cos(\alpha)$$

$$\cos\left(\alpha+\frac{\pi}{2}\right)=-\sin(\alpha).$$

$$\alpha_1=\frac{\pi}{2}-\theta t.$$

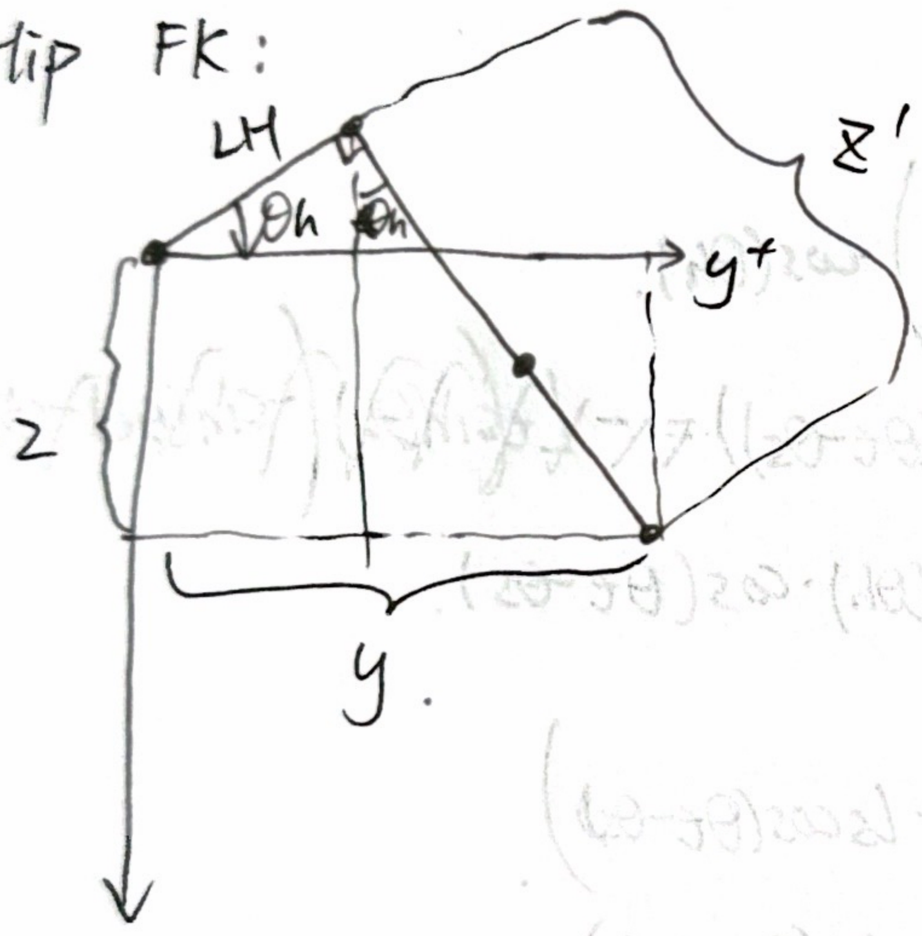
$$\begin{cases}\beta_1=\pi-\theta s-\alpha_1\\ \beta_1=\theta t-\theta s+\frac{\pi}{2}\end{cases}$$

$$x=Lt\sin(\theta t)-Ls\cos\left(\theta t-\theta s+\frac{\pi}{2}\right)$$

$$\Downarrow$$

$$\begin{cases}x=Lt\sin(\theta t)+Ls\sin(\theta t-\theta s)\\ z'=Lt\cos(\theta t)+Ls\cos(\theta t-\theta s)\end{cases}$$

2. Hip FK:



$$\begin{cases}y=Lh\cos(\theta h)+z'\sin(\theta h)\\ z=-Lh\sin(\theta h)+z'\cos(\theta h)\end{cases}$$

z+

$$P(x,y,z)=\begin{bmatrix}Lt\sin(\theta t)+Ls\sin(\theta t-\theta s)\\ Lh\cos(\theta h)+z'\sin(\theta h)\\ -Lh\sin(\theta h)+z'\cos(\theta h)\end{bmatrix}=\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\vec{P}.$$

$$\text{Jacobian} = \begin{bmatrix} \frac{\partial x}{\partial \theta h} & \frac{\partial x}{\partial \theta t} & \frac{\partial x}{\partial \theta s} \\ \frac{\partial y}{\partial \theta h} & \frac{\partial y}{\partial \theta t} & \frac{\partial y}{\partial \theta s} \\ \frac{\partial z}{\partial \theta h} & \frac{\partial z}{\partial \theta t} & \frac{\partial z}{\partial \theta s} \end{bmatrix}.$$

$$(\sin(t))' = \cos(t)$$
$$(\cos(t))' = -\sin(t)$$

$$\begin{cases} x = Lt\sin(\theta t) + Ls\sin(\theta t - \theta s) \\ y = Lh\cos(\theta h) + z'\sin(\theta h) = Lh\cos(\theta h) + \left(Lt\cos(\theta t)\cdot\sin(\theta h) + Ls(\cos(\theta t - \theta s)\cdot\sin(\theta h)\right). \\ z = -Lh\sin(\theta h) + z'\cos(\theta h) = -Lh\sin(\theta h) + \left(Lt\cos(\theta t)\cdot\cos(\theta h) + Ls\cos(\theta t - \theta s)\cdot\cos(\theta h)\right). \\ z = Lt\cos(\theta t) + Ls\cos(\theta t - \theta s). \end{cases}$$

$$\begin{cases} \frac{\partial x}{\partial \theta h} = 0. \\ \frac{\partial x}{\partial \theta t} = Lt\cos(\theta t) + Ls\cos(\theta t - \theta s). \\ \frac{\partial x}{\partial \theta s} = Ls\cos(\theta t) - Ls\cos(\theta t - \theta s). \end{cases}$$

$$\begin{cases} \frac{\partial y}{\partial \theta h} = -Lh\sin(\theta h) + \left(Lt\cos(\theta t) + Ls\cos(\theta t - \theta s)\right)\cdot\cos(\theta h). \\ \frac{\partial y}{\partial \theta t} = Lt\sin(\theta h)(-\sin(\theta t)) + Ls\sin(\theta h)(-\sin(\theta t - \theta s)) \\ \frac{\partial y}{\partial \theta s} = Ls\sin(\theta h)(-1)\cdot(-\cos(\theta t - \theta s)) = Ls\sin(\theta h)\cdot\cos(\theta t - \theta s). \end{cases}$$

$$\begin{cases} \frac{\partial z}{\partial \theta h} = -Lh\cdot\cos(\theta h) - \sin(\theta h)\cdot\left(Lt\cos(\theta t) + Ls\cos(\theta t - \theta s)\right). \\ \frac{\partial z}{\partial \theta t} = Lt\cos(\theta h)(-\sin(\theta t)) + Ls\cos(\theta h)\cdot(-\sin(\theta t - \theta s)). \\ \frac{\partial z}{\partial \theta s} = Ls\cos(\theta h)(-1)(-1)\cdot\sin(\theta t - \theta s) = Ls\cos(\theta h)\cdot\sin(\theta t - \theta s). \end{cases}$$

Jacobian =

$$\begin{bmatrix} 0 & L_t\cos(\theta_t)+L_s\cos(\theta_t-\theta_s) & L_s\cos(\theta_t)-L_s\cos(\theta_t-\theta_s) \\ -L_h\sin(\theta_h)+\cos(\theta_h)\cdot\left(L_t\cos(\theta_t)+L_s\cos(\theta_t-\theta_s)\right) & \sin(\theta_h)\cdot\left(-L_t\sin(\theta_t)-L_s\sin(\theta_t-\theta_s)\right) & L_s\sin(\theta_h)\cdot\cos(\theta_t-\theta_s) \\ -L_h\cos(\theta_h)-\sin(\theta_h)\cdot\left(L_t\cos(\theta_t)+L_s\cos(\theta_t-\theta_s)\right) & \cos(\theta_h)\left(-L_t\sin(\theta_t)-L_s\sin(\theta_t-\theta_s)\right) & L_s\cos(\theta_h)\sin(\theta_t-\theta_s) \end{bmatrix}$$